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Dynamic Logits Adjustment and Exploration for Test-Time Adaptation in Vision Language Models

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Abstract

Existing Test-Time Adaptation (TTA) methods for Vision-Language Models (VLMs), focusing on designing efficient adaptation parameters (e.g., prompts or residual prototypes), predominantly rely on high-confidence samples obtained via entropy-based filtering. However, this prevailing paradigm implicitly inherits the VLM's class-wise prediction biases and leads to insufficient coverage of the test distribution, rendering the adaptation process biased and insufficiently exploratory. To overcome these limitations, we propose Dynamic Logits Adjustment and Exploration (DLAE), a novel framework that integrates Dynamic Logit Adjustment (DLA) with a Consistency-Guided Exploratory Cache (CGEC). DLA dynamically recalibrates model logits based on test prediction statistics, thereby mitigating class-wise prediction inconsistencies. Different from traditional cache mechanisms, our CGEC actively identifies additional samples near decision boundaries whose predicted labels are sensitive to the logit adjustment, thereby exploring beyond only high-confidence samples. By enforcing semantic and temporal consistency, the cache preserves the reliability of selected samples while enabling cautious yet effective exploration of low-confidence regions, ultimately yielding stable and reliable adaptation. Extensive experiments across multiple vision-language benchmarks demonstrate that our approach consistently surpasses state-of-the-art TTA methods, showing superior stability, adaptability, and generalization. Code is available here.

1. Introduction

Vision Language Models (VLMs), such as CLIP [27] and ALIGN [15], are now the standard foundation for numerous visual perception tasks. In practice, however, deployment always faces a distribution shift between the general-purpose pretraining corpus and the target environment. Test-Time Adaptation (TTA) [17, 31, 43, 45, 46] provides an attractive remedy because it adapts a frozen or lightly updated model at inference without target labels and without costly offline retraining. Recent work on sequential TTA for VLMs [21, 43, 46], especially in the online streaming setting, can be broadly grouped into two families. The first family focuses on prompt-based adaptation, where backbone parameters are frozen and only continuous textual prompts are optimized to better align text features with the test stream [8, 29]. The second family employs cache-based mechanisms that store historical featurelabel pairs or class prototypes [17, 43, 45, 46] and use retrieval or prototype guidance to stabilize predictions. Within this family, recent work [43] further exploits the cache to drive prototype op-

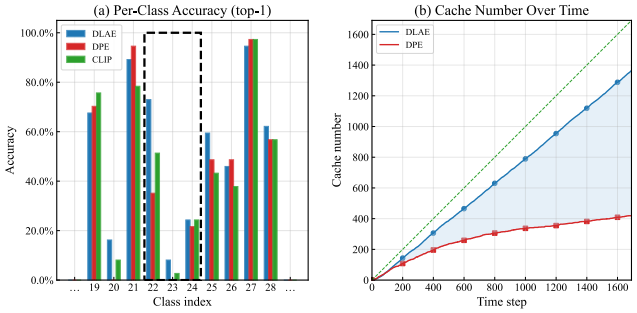


Figure 1. (a) Per-class accuracy (top-1) on the target stream for CLIP in zero-shot mode, DPE as a cache-based TTA method, and our DLAE. CLIP exhibits large per-class variance. The black dashed box highlights classes where DPE underperforms and even falls below the zero-shot baseline, while our method achieves more consistent accuracy across classes. (b) Number of distinct target samples entering the cache over the test stream as the time step increases. Both DPE and DLAE use caches with the same fixed capacity and update them throughout the stream. However, the minimum-entropy replacement policy of DPE causes the cache to be dominated by low-entropy samples, preventing many diverse samples from entering. In contrast, our DLAE enables a much larger and more informative set of samples to be included within the same fixed-capacity cache.

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timization in the embedding space by computing residual prototypes from high-confidence test samples, typically selected via entropy-based filtering.

Despite this progress, two systemic limitations persist. First, large-scale pretraining on heterogeneous web corpora induces class-wise prediction biases in VLMs, which are further amplified by standard TTA pipelines. Most existing methods rely on entropy-based filtering to select "reliable" target samples. This design implicitly trusts the initial class preferences of the model: early predictions on relatively easy classes pass the confidence threshold more often and dominate the pseudo-labels, so most predicted labels fall into only a few categories. This imbalance gradually steers adaptation away from where improvement is most needed and can eventually lead to pseudo-label collapse. Second, cache designs commonly prioritize high-confidence samples and under-explore the target distribution. With a fixed cache capacity and replacement strategy, only a small portion of the stream is ever admitted. If admission is tied almost exclusively to high confidence, the cache becomes a tiny anchor set skewed toward easy, already well-modeled regions. The resulting prototypes or retrieval statistics can be poorly representative of the full target distribution, especially in low-confidence regions, which further reinforces the model bias described above.

We address these challenges with *Dynamic Logits Adjustment and Exploration* (DLAE), a framework that couples *Dynamic Logit Adjustment* (DLA) with a *Consistency-Guided Exploratory Cache* (CGEC). DLA maintains an online estimate of the test-time label distribution via prediction statistics and applies class-specific logit shifts to recalibrate model outputs. By counteracting over- and under-predicted classes at the logit level, DLA reduces class-wise inconsistencies before pseudo-labels are committed. This recalibration not only leads to more balanced treatment of rare or difficult classes, but also naturally reveals samples near the decision boundary, precisely those whose predicted labels are most sensitive to the adjustment. Leveraging this property, CGEC departs from prior cache-based approaches that exclusively select high-confidence predictions. Instead, it actively incorporates boundary samples whose labels change under DLA, thereby targeting examples in low-confidence and previously under-explored regions of the stream. To control noise during exploration, cache admission is governed by two complementary consistency checks: (1) semantic consistency, requiring that the labels assigned before and after DLA remain close in the text-embedding space; and (2) Temporal consistency, requiring that a samples feature representation becomes closer to the current text prototype than to the initial prototype. Samples that satisfy both criteria are likely to be reliable while still promoting coverage beyond the high-confidence regime.

In summary, our framework delivers stable and effective test-time learning for VLM-based classification under distribution shift. It reduces class-wise bias, unlocks controlled exploration, and avoids the collapse induced by confidence-only selection. Our main contributions are as follows:

- We reveal that class-wise prediction biases in large-scale vision-language models, amplified by confidence-based sample selection, skew pseudo-labels, hinder exploration, and can lead to collapse in sequential test-time adaptation.
- We propose a TTA framework called Dynamic Logits Adjustment and Exploration (DLAE), which combines DLA for class-specific logit recalibration from online prediction statistics with a CGEC that, guided by DLA, admits additional samples near decision boundaries beyond only high-confidence ones, thereby improving coverage of the test distribution.
- We conduct extensive experiments on two challenging benchmarks and demonstrate that DLAE consistently outperforms state-of-the-art test-time adaptation methods for vision-language models, achieving more accurate, stable, and robust performance under distribution shift.

2. Related Work

2.1. Vision-Language Models

Vision-Language Models (VLMs) pre-trained on web-scale image-text pairs, such as CLIP [27] and ALIGN [15], learn transferable visual representations from natural-language supervision and support the pre-train, fine-tune paradigm. These models enable competitive zero- and few-shot generalization and serve as strong backbones for downstream tasks, including image recognition [11, 22, 27], object detection [38, 39], and video understanding [18, 23, 36].

To adapt these powerful pre-trained models to specific downstream tasks, a spectrum of parameter-efficient strategies has been explored. Prompt-based tuning methods, *e.g.*, CoOp [49] and CoCoOp [50], optimize textual prompts to align class semantics with visual features better and consistently improve few-shot performance. More recent works [20, 37] introduced efficient adaptation strategies by leveraging feature memory constructed from a small set of labeled samples.

2.2. Test-Time Adaptation in VLMs

Test-time adaptation [2, 32–34, 44] aims to enhance the adaptability of VLMs to target domains using only unlabeled test samples. Recent VLM-TTA methods can be broadly categorized into three paradigms: prompt-based adaptation, cache-based adaptation, and statistics/distribution-based adaptation.

Prompt-based methods [8, 29] optimize continuous textual prompt embeddings at test time to bridge the domain gap without updating the backbone. TPT [29] first intro-

duced test-time prompt tuning by enforcing prediction consistency across augmented views, while DiffTPT [8] improves robustness with diffusion-based data augmentations to better handle diverse target variations.

Cache-based methods [3, 43, 46] leverage memory mechanisms to store and exploit information from historical test samples. For example, DPE [43] jointly evolves visual and textual prototypes during inference to capture domain-specific semantics dynamically. Other approaches further extend this direction by maintaining multiple cache pools to store diverse and reliable test examples, improving stability under distribution shifts [3, 46].

Beyond explicit prompts and feature caches, distribution-based methods adapt predictions by updating calibration states or modeling target distributions from test-stream evidence. These methods can be further grouped into batch-based and streaming schemes. Batch-based approaches exploit batch-level structure, e.g., similarity-based graph models for label propagation or batch-wise distribution modeling to refine predictions and calibrate decision boundaries [16, 41, 42, 48]. In contrast, streaming approaches continuously update adaptation states along the test stream, such as caching feature statistics [9], modeling distributional states for TTA in VLMs [10], or performing online distribution updates and alignment [5, 47].

However, class-wise prediction bias can cause certain categories to be systematically under-predicted on the target stream. As a consequence, prior test-time adaptation methods can be driven by skewed pseudo-labels or misestimated running statistics, resulting in a self-reinforcing error loop that is difficult to correct, especially when the base model is weak on the target domain. To address this failure mode, we perform streaming logit-space debiasing to surface hard target samples whose predictions are sensitive to the debiasing operation, and then leverage a consistency-guided exploratory cache to reliably mine and learn from these challenging regions.

3. Method

3.1. Preliminaries

Zero-Shot CLIP. CLIP [27] aligns images and text in a shared d-dimensional embedding space via pre-trained visual and textual encoders $\mathcal{E}_v(\cdot)$ and $\mathcal{E}_t(\cdot)$. For a C-class zero-shot classification task, an input image $\mathbf{X}_{\text{test}}$ is mapped to $\mathbf{f}_v = \mathcal{E}_v(\mathbf{X}_{\text{test}})$ and compared with text embeddings $\mathbf{t}_c = \mathcal{E}_t(\mathbf{T}_c)$ of class prompts.For simplicity, we omit the notation indicating ℓ_2-normalization of the embeddings, which is applied in practice. The zero-shot probabilities are computed via softmax with the temperature τ:

$$P_{clip}(y = c \mid \mathbf{X}_{\text{test}}) = \frac{\exp(\mathbf{f}_v^\top \mathbf{t}_c/\tau)}{\sum_{c'=1}^{C} \exp(\mathbf{f}_v^\top \mathbf{t}_{c'}/\tau)}, \quad (1)$$

Cache-Based Prototype Learning. Following [43], a per-class cache $\{\mathcal{P}_c\}_{c=1}^{C}$ is maintained as a priority queue that preferentially retains entries with low predictive entropy. Each cached element consists of a pair $[\mathbf{f}_v, h]$, where $h = \mathcal{H}(P_{clip}(\mathbf{X}_{\text{test}}))$ denotes the predictive Shannon entropy. The CLIP prediction $\hat{y}_{clip} = \arg\max_c P_{clip}(\mathbf{X}_{\text{test}})$ is used as a pseudo-label and the pair $[\mathbf{f}_v, h]$ is inserted into corresponding cache $\mathcal{P}_{\hat{y}_{\text{CLIP}}}$. Given the cache, the visual prototype for class c is computed as:

$$\mathbf{v}_c = \frac{1}{|\mathcal{P}_c|} \sum_{[\mathbf{f}_v,\, h] \in \mathcal{P}_c} \mathbf{f}_v. \quad (2)$$

To enable test-time adaptation, we further associate each class c with a learnable text residual $\Delta\mathbf{t}_c \in \mathbb{R}^d$ and a learnable visual residual $\Delta\mathbf{v}_c \in \mathbb{R}^d$. These residual parameters are initialized to zero and optimized via gradient descent during test-time adaptation, while the original CLIP text embeddings and cache-based prototypes serve as fixed anchors. The adapted text and visual prototypes are obtained by adding the residuals and then applying ℓ_2 normalization:

$$\mathbf{t}_c \leftarrow \frac{\mathbf{t}_c + \Delta\mathbf{t}_c}{\|\mathbf{t}_c + \Delta\mathbf{t}_c\|_2}, \quad \mathbf{v}_c \leftarrow \frac{\mathbf{v}_c + \Delta\mathbf{v}_c}{\|\mathbf{v}_c + \Delta\mathbf{v}_c\|_2}. \quad (3)$$

After this update, we reuse the notation $\mathbf{t}_c$ and $\mathbf{v}_c$ to denote the adapted text and visual prototypes, respectively. The overall objective $\mathcal{L}_{all}$ consists of a confidence-aware calibration loss $\mathcal{L}_{conf}$ and a symmetric cross-modal alignment loss $\mathcal{L}_{align}$:

$$\mathcal{L}_{all} = \mathcal{L}_{conf} + \lambda \mathcal{L}_{align}, \quad (4)$$

$$\mathcal{L}_{conf} = \mathcal{H}\left(\frac{\exp\left([\mathbf{f}_v^\top \mathbf{t}_c + \mathcal{A}(\mathbf{f}_v^\top \mathbf{v}_c)]/\tau\right)}{\sum_{c'} \exp\left([\mathbf{f}_v^\top \mathbf{t}_{c'} + \mathcal{A}(\mathbf{f}_v^\top \mathbf{v}_{c'})]/\tau\right)} \right), \quad (5)$$

$$\mathcal{L}_{align} = \frac{1}{C} \sum_{c=1}^{C} \left(-\log \frac{\exp(\mathbf{t}_c^\top \mathbf{v}_c)}{\sum_{c'}^{C} \exp(\mathbf{t}_c^\top \mathbf{v}_{c'})} - \log \frac{\exp(\mathbf{t}_c^\top \mathbf{v}_c)}{\sum_{c'}^{C} \exp(\mathbf{t}_{c'}^\top \mathbf{v}_c)} \right). \quad (6)$$

Here, λ controls the trade-off between the two loss terms, τ denotes the temperature, and $\mathcal{A}(\cdot)$ is the affinity modulation function inherited from [43]. During inference, the final score s_{pre}^c for class c is computed by combining the semantic similarity with the affinity-modulated visual cue:

$$s_{pre}^c = \mathbf{f}_v^\top \mathbf{t}_c + \mathcal{A}(\mathbf{f}_v^\top \mathbf{v}_c). \quad (7)$$

3.2. Dynamic Logits Adjustment (DLA)

As shown in Fig. 1 (a), existing TTA methods suffer from pronounced class-dependent bias. Some categories are

recognized with consistently high accuracy, whereas others suffer from severe underperformance and even fall below the CLIP zero-shot baseline, as indicated by the black dashed region. This phenomenon reflects the models inherently uneven recognition capability across categories, which becomes amplified during test-time self-training. High-confidence classes are repeatedly reinforced, while rare or challenging ones receive few or no updates.

To mitigate this issue, we propose Dynamic Logits Adjustment (DLA), an online calibration mechanism that adjusts prediction logits during test-time adaptation. DLA operates entirely on a per-sample basis, requiring neither retraining nor access to source data. Instead, it leverages lightweight, continuously updated per-class statistics derived from incoming pseudo-labels. Specifically, for each class c, DLA maintains two running estimates: a counter $\mathbf{n} \in \mathbb{R}^C$, where $\mathbf{n}[c]$ accumulates the number of times class c has been assigned as a pseudo-label to estimate the empirical class distribution (initialized as 1); and a running average confidence $\boldsymbol{\mu} \in \mathbb{R}^C$, where $\boldsymbol{\mu}[c]$ tracks the historical mean confidence of predictions assigned to class c as a reliability indicator (initialized as 0).

Given a test image with visual feature $\mathbf{f}_v$, we first compute the raw CLIP logits $\mathbf{s}^c_{clip} = \mathbf{f}_v^\top \mathbf{t}_c$ and obtain the initial pseudo-label $\hat{y}_{\text{CLIP}}$. The empirical class probability is estimated as:

$$\hat{p}(c) = \frac{\mathbf{n}[c]}{N}, \quad \text{where } N = \sum_{c'=1}^{C} \mathbf{n}[c']. \tag{8}$$

The core of DLA is a balancing function $B(c)$ that modulates class logits based on both the observed frequency and the reliability of each class:

$$B(c) = \exp\big(-\alpha \cdot \hat{p}(c) \cdot (1-d)\big), \tag{9}$$

where $d = P_{clip}(\hat{y}_{clip}) - \boldsymbol{\mu}[\hat{y}_{clip}]$ quantifies the deviation of the current prediction confidence from the historical average for the predicted class, and α is a hyperparameter controlling the adjustment strength.

This design yields three key behaviors. First, when a frequently predicted class receives a confidence lower than its historical norm ($d < 0$), the factor $B(c)$ becomes small, attenuating potentially overconfident but unreliable predictions. When the confidence aligns with or exceeds the historical mean ($d \geq 0$), $B(c)$ remains close to one, maintaining stable predictions. Third, for rare classes, the small value of $\hat{p}(c)$ keeps $B(c)$ near one, ensuring that tail-class predictions are not overly penalized and can effectively contribute to adaptation. The final adjusted logit s^c_{DLA} for class c is:

$$s^c_{\text{DLA}} = \mathbf{s}^c_{clip} \cdot B(c). \tag{10}$$

After each prediction, the class counter $\mathbf{n}$ and mean confidence $\boldsymbol{\mu}$ are updated incrementally using the new pseudo-label and confidence score, ensuring that both statistics reflect the evolving test-time distribution. Detailed update rules are provided in the Appendix.

3.3. Consistency-Guided Exploratory Cache

While DLA balances prediction confidence across classes, Fig. 1 (b) reveals a complementary limitation in cache-based TTA methods. Existing caching strategies, such as DPE, quickly saturate the fixed-capacity cache, and the number of distinct stored samples plateaus early in the adaptation process. This behavior implies that the cache repeatedly absorbs a narrow subset of easy, high-confidence instances and therefore provides limited coverage of the broader test distribution. As a result, potentially informative yet underrepresented samples remain excluded from the adaptation loop.

To address this limitation, we revisit the effect of DLA on prediction dynamics. Beyond producing calibrated logits, DLA naturally exposes a set of informative candidates whose predictions flip after adjustment (i.e., $\hat{y}_{clip} \neq \hat{y}_{\text{DLA}}$). These flipped samples are often discarded by existing cache-based methods due to their higher uncertainty, yet they often reside near decision boundaries where the model is most in need of new guidance. Although directly caching all such samples would introduce noise and destabilize adaptation, many of them contain critical semantic signals that can help mitigate class bias and prevent pseudo-label collapse when handled properly.

To exploit these samples safely, we introduce the Consistency-Guided Exploratory Cache (CGEC). Rather than replacing the high-confidence samples typically stored in cache-based methods, CGEC supplements them with flipped samples under a semantic consistency constraint and all samples under a complementary temporal one.

Semantic Consistency Filter (SCF) The SCF ensures that the flipped samples admitted into the cache remain semantically reliable when used to supplement the existing high-confidence instances. To enforce semantic consistency, we require that the labels assigned before and after DLA stay close in the text-embedding space. For example, a flip from "dog" to "wolf" is semantically coherent and should tighten prediction, whereas a flip from "dog" to "airplane" signals inconsistency and should not reduce uncertainty. We quantify this consistency using the cosine similarity between the text embeddings of the original and refined predictions, and use it to modulate the samples entropy:

$$h \leftarrow h \cdot \exp(-\beta \cdot \cos\left(\mathbf{t}_{\hat{y}_{\text{clip}}}, \mathbf{t}_{\hat{y}_{\text{DLA}}}\right)). \tag{11}$$

This formulation reduces entropy for semantically consistent flips, making them more likely to be retained in the cache as informative boundary samples.

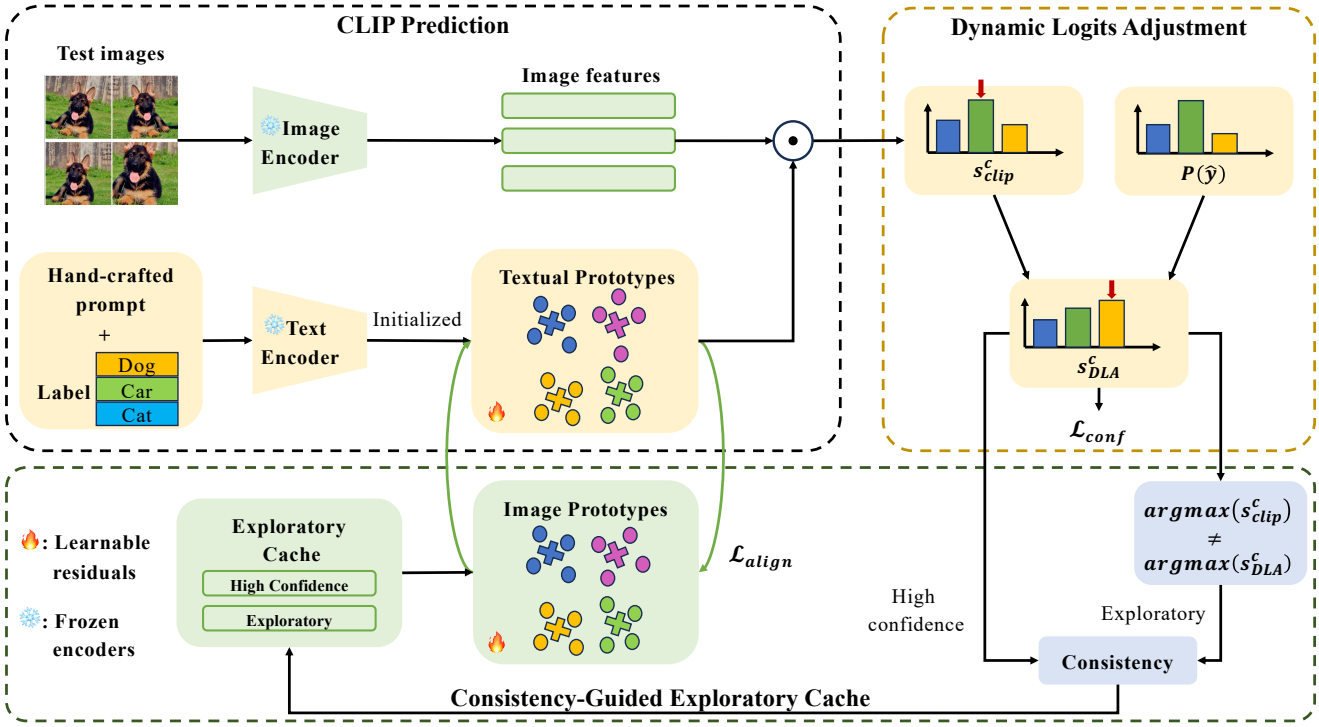


Figure 2. An illustration of the proposed DLAE framework. The framework consists of two main components: Dynamic Logits Adjustment (DLA) and a Consistency-Guided Exploratory Cache (CGEC). In the CLIP prediction stage, image and text encoders extract image features and textual prototypes initialized from hand-crafted prompts. DLA recalibrates logits using the estimated target prediction distribution to reduce prediction inconsistency and confidence bias. CGEC stores test samples while focusing more on those whose predicted labels change before and after DLA, encouraging exploration in low-confidence regions. Guided by semantic and temporal consistency, the cache updates image and textual prototypes through learnable residuals, enabling the model to explore low-confidence regions while maintaining stable and consistent adaptation throughout the test stream.

Temporal Consistency Filter (TCF) The TCF evaluates all samples admitted into the cache, including both high-confidence samples and flipped samples. Its purpose is to ensure that their feature representations evolve coherently with the model during adaptation. For each cached instance, we track whether its original visual feature $\mathbf{f}_v$ becomes less aligned with the evolving text embedding of its assigned class $\hat{y}_{\text{DLA}}$ over time. Specifically, let $\mathbf{t}^\top_{\hat{y}_{\text{DLA}}}[0]$ denote the initial text embedding of class $\hat{y}_{\text{DLA}}$ from the model's predefined initial state, and $\mathbf{t}^\top_{\hat{y}_{\text{DLA}}}[\text{now}]$ represent the current text embedding for the same class. Temporal inconsistency is detected when

$$\mathbf{t}^\top_{\hat{y}_{\text{DLA}}}[0]\mathbf{f}_v \;>\; \mathbf{t}^\top_{\hat{y}_{\text{DLA}}}[\text{now}]\mathbf{f}_v. \tag{12}$$

When this condition is met, we penalize the sample by increasing its modulated entropy:

$$h \leftarrow h \cdot \exp\left(\eta\left(i_{\text{now}} - i_{\text{entry}}\right)\right), \tag{13}$$

where i_{now} denotes the current training time step and i_{entry} represents the training time step when the sample was first added to the cache. This mechanism ensures that the cache retains only those boundary samples that remain temporally consistent with the evolved model, effectively filtering out those that become outliers due to representation drift.

By jointly applying SCF and TCF, CGEC modulates sample entropy according to semantic and temporal consistency, so that reliable samples, especially flipped ones, are promoted in a low-entropy priority queue that broadens exploration around decision boundaries.

3.4. Overall Model

As shown in Fig. 2, our full framework integrates the DLA-calibrated semantics with the CGEC-refined visual prototypes into a unified prediction model. For each class c, we replace the raw CLIP logit s^c_{clip} with the calibrated score s^c_{DLA} and construct the visual prototype $\mathbf{v}_c$ from features stored in CGEC. The final prediction score is:

$$s^c_{\text{DLAE}} = s^c_{\text{DLA}} + \mathcal{A}(\mathbf{f}_v^\top \mathbf{v}_c), \tag{14}$$

which jointly reflects the calibrated semantic evidence and the affinity-modulated visual prototype.

Training follows the alignment-confidence objective in Eq. 4 with two modifications. First, the semantic similarity

Table 1. Top-1 accuracy (in %) on cross-datasets generalization. For all evaluated methods, we report the results on both ResNet-50 and ViT-B/16 visual backbones of CLIP.

Method	Caltech	DTD	Cars	EuroSAT	Aircraft	Flowers	Pets	UCF101	Food101	SUN397	Average
CLIP-RN50 [27]	85.88	40.37	55.70	23.69	15.66	61.75	83.57	58.84	73.97	58.80	55.82
CoOp [49]	86.53	37.29	55.32	26.20	15.12	61.55	87.00	59.05	75.59	58.15	56.18
TPT [29]	87.02	40.84	58.46	28.33	17.58	62.69	84.49	60.82	74.88	61.46	57.66
DiffTPT [8]	86.89	40.72	60.71	41.04	17.60	63.53	83.40	62.67	**79.21**	62.72	59.85
TDA [17]	89.70	43.74	57.78	42.11	17.61	68.74	86.18	64.18	77.75	62.53	61.03
BCA [51]	89.70	48.58	58.13	42.12	19.89	66.30	85.58	63.51	77.19	63.38	61.44
DPE [43]	90.83	50.18	59.26	41.67	19.80	67.60	85.97	61.98	77.83	64.23	61.93
SCA [9]	**91.03**	53.31	59.73	43.33	20.82	**71.09**	86.15	**66.40**	76.51	64.50	63.29
DMN-ZS [46]	90.14	50.41	60.02	48.72	22.77	67.93	86.78	65.34	76.70	64.39	63.71
DLAE (Ours)	90.91	**55.56**	**61.98**	**50.04**	**23.34**	69.63	**87.68**	64.76	78.76	**65.43**	**64.81**
CLIP-ViT-B/16 [27]	93.35	44.27	65.48	42.01	23.67	67.44	88.25	65.13	83.65	62.59	63.58
CoOp [49]	93.70	41.92	64.51	46.39	18.47	68.71	89.14	66.55	85.30	64.15	63.88
TPT [29]	94.16	47.75	66.87	42.44	24.78	68.98	87.79	68.04	84.67	65.50	65.10
DiffTPT [8]	92.49	47.00	67.01	43.13	25.60	70.10	88.22	62.67	**87.23**	65.74	65.47
TPS [31]	95.09	50.47	69.06	44.48	26.34	71.54	87.35	71.00	85.23	68.98	66.96
BoostAdapter [45]	94.77	45.69	69.30	61.22	27.45	71.66	89.51	71.93	87.17	68.09	68.68
TDA [17]	94.24	47.40	67.28	58.00	23.91	71.42	88.63	70.66	86.14	67.62	67.53
BCA [51]	94.69	53.49	66.86	56.63	28.59	73.12	90.43	67.59	85.97	68.41	68.59
DMN-ZS [46]	**95.38**	55.85	67.96	59.43	30.03	74.49	92.04	72.51	85.08	70.18	70.30
DPE [43]	94.81	54.20	67.31	55.79	28.95	75.07	91.14	70.44	86.17	70.07	69.40
SCA [9]	94.85	57.09	68.49	57.16	28.50	76.09	91.44	73.43	86.09	70.27	70.34
DLAE (Ours)	95.33	**58.86**	**69.99**	**61.30**	**32.58**	**77.50**	**91.88**	**73.46**	87.05	**70.83**	**71.88**

term in $\mathcal{L}_{\text{conf}}$ is replaced by s^c_{DLAE}:

$$\mathcal{L}^{\text{DLAE}}_{\text{conf}} = \mathcal{H}\left(\frac{\exp(s^c_{\text{DLAE}}/\tau)}{\sum_{c'}\exp(s^{c'}_{\text{DLAE}}/\tau)}\right) \tag{15}$$

ensuring that calibration is applied directly to the final combined logits. Second, each prototype $\mathbf{v}_c$ is computed from the CGEC, while the remaining components of $\mathcal{L}_{\text{align}}$ and the optimization pipeline remain identical to Sec. 3.1. At test time, prediction $\hat{y}_{\text{DLAE}}$ is obtained by selecting the class with the highest combined score:

$$\hat{y}_{\text{DLAE}} = \arg\max_c s^c_{\text{DLAE}}. \tag{16}$$

4. Experiments

4.1. Experimental Setup

Benchmarks. We evaluate our method's effectiveness and robustness on two prevalent test time adaptation (TTA) benchmarks: the Cross Domain (CD) benchmark and the Out of Distribution (OOD) benchmark. The CD benchmark assesses cross-domain image classification adaptability and includes ten non-overlapping visual domain datasets: Aircraft [24], Caltech101 [7], Cars [19], DTD [4], EuroSAT [12], Flowers102 [25], Food101 [1], Pets [26], SUN397 [40], and UCF101 [30]. It measures performance under significant semantic and distributional shifts. Each dataset corresponds to a distinct domain and label space, covering fine-grained object recognition and textures, which closely mimics real-world deployments where the target classes and visual statistics can differ substantially. OOD benchmark is built around the ImageNet dataset [6], which we treat as the in-distribution reference, together with four challenging derived variants: ImageNet-A [14], ImageNet-V2 [28], ImageNet-R [13], and ImageNet-S [35]. These datasets contain natural adversarial examples and pronounced distribution shifts.

4.2. Implementation Details

We conduct all experiments on pretrained CLIP [27] models with ResNet-50 and ViT-B/16 vision encoders, and run all evaluations on a single 24 GB NVIDIA RTX 3090 GPU. Test-time adaptation is performed in a single-image setting with a test-time batch size of 1. Following TPT [29], we apply random resized cropping to generate up to 63 augmented views for each test image to enhance robustness. The cache capacity is set to 3, matching the configuration of DPE [43] to ensure a fair comparison under identical memory constraints. The impact of these hyperparameters is further analyzed in the ablation study section.

4.3. Comparisons with State-of-the-Art

We compare our method with representative CLIP-based TTA approaches [2, 32–34, 44]. These baselines cover prompt-based methods such as TPT [29] and DiffTPT [8], cache-based methods including DMN-ZS [46], TDA [17], and DPE [43], as well as distribution-based methods that

Table 2. Top-1 accuracy (in %) on robustness to natural distribution shifts. For all evaluated methods, we report the results on both ResNet-50 and ViT-B/16 visual backbones of CLIP.

Method	ImageNet	ImageNet-A	ImageNet-V2	ImageNet-R	ImageNet-S	Average	OOD Average
CLIP-RN50 [27]	58.16	21.83	51.41	56.15	33.37	44.18	40.69
CoOp [49]	63.33	23.06	55.40	56.60	34.67	46.61	42.43
TPT [29]	60.74	26.67	54.70	59.11	35.09	47.26	43.89
DiffTPT [8]	60.80	31.06	55.80	58.80	37.10	48.71	45.69
TPS [31]	61.47	30.48	54.96	62.87	37.14	49.38	46.36
TDA [17]	61.35	30.29	55.54	62.58	38.12	49.58	46.63
BCA [51]	61.81	30.35	56.58	62.89	38.08	49.94	46.98
DMN-ZS [46]	63.87	28.57	56.12	61.44	39.84	49.97	46.49
SCA [9]	63.11	30.64	56.51	62.85	39.33	50.49	47.33
DPE [43]	63.41	30.15	56.72	**63.72**	40.03	50.81	47.66
DLAE (Ours)	**64.10**	**34.80**	**57.13**	63.63	**40.68**	**52.06**	**49.06**
CLIP-ViT-B/16 [27]	66.73	47.87	60.86	73.98	46.09	59.11	57.20
CoOp [49]	71.51	49.71	64.20	75.21	47.99	61.72	59.28
DiffTPT [8]	70.30	55.68	65.10	75.00	46.80	62.28	60.52
TPT [29]	68.98	54.77	63.45	77.06	47.94	62.44	60.81
TDA [17]	69.51	60.11	64.67	80.24	50.54	65.01	63.89
TPS [31]	70.19	60.08	64.73	80.27	49.95	65.04	63.07
BCA [51]	70.22	61.14	64.90	80.72	50.87	65.37	64.16
DMN-ZS [46]	72.25	58.28	65.17	78.55	**53.20**	65.49	63.80
DPE [43]	71.91	59.63	65.44	80.40	52.26	65.93	64.43
SCA [9]	71.75	60.33	65.38	**80.85**	52.50	66.16	64.77
DLAE (Ours)	**72.41**	**64.06**	**65.50**	80.43	53.06	**67.09**	**65.76**

adapt predictions from test-time evidence [5, 9]. All baselines are evaluated under the same CLIP backbone and evaluation protocol [27].

Cross-datasets generalization. Tab. 1 summarizes the results on ten diverse downstream datasets. Using the CLIP-RN50 backbone, DLAE achieves 64.81% average accuracy, substantially higher than the CLIP baseline (55.82%) and outperforming the strongest competing cache-based approaches, including DPE (61.93%) and DMN-ZS (63.71%). With the stronger CLIP-ViT-B/16 backbone, DLAE further boosts the average accuracy to 71.88%, improving upon vanilla CLIP (63.58%) as well as advanced TTA methods including DPE (69.40%) and DMN-ZS (70.30%). Beyond aggregate performance, DLAE achieves the best accuracy on the majority of evaluated datasets. Its advantages are especially notable on texture recognition (DTD), remote sensing (EuroSAT), and large-scale scene understanding (SUN397). These results highlight not only the robustness of DLAE under domain shift but also its strong scalability as the visual backbone becomes more expressive.

Natural distribution shifts. Tab. 2 evaluates robustness on ImageNet and its natural distribution shift variants. With the ResNet-50 backbone, DLAE achieves the highest OOD average of 49.06% and overall average of 52.06%, outperforming competitive methods such as DPE (47.66% and 50.81%). The improvement on ImageNet-A is especially significant, where DLAE reaches 34.80%, well above prior TTA approaches. With the ViT-B/16 backbone, DLAE further improves robustness, achieving an OOD average of 65.76% and an overall average of 67.09%. Both metrics exceed those of the strongest baseline DPE. It delivers state-of-the-art performance on ImageNet, ImageNet-A, and ImageNet-V2, while remaining highly competitive on ImageNet-R and ImageNet-Sketch, demonstrating strong robustness to natural distribution shifts across both convolutional and transformer backbones.

4.4. Ablation Studies

In this section, we present a comprehensive ablation study of the proposed DLAE using CLIP [27] with a ViT-B/16 backbone. We isolate the contributions of each component and analyze the sensitivity of key hyperparameters to better understand the behavior of our framework.

Ablation on the components of DLAE. Tab. 3 (a) summarizes the effects of Dynamic Logits Adjustment (DLA) and the Consistency-Guided Exploratory Cache (CGEC). Removing both modules collapses the method to the baseline, yielding an accuracy of 69.40%. Introducing DLA alone increases performance to 70.63%, indicating that dynamically adjusting logits and encouraging exploration effectively mitigates model bias under distribution shifts. Adding CGEC on top of DLA further improves the accu-

Table 3. Ablation study on components of (a) Dynamic Logits Adjustment and Exploration, (b) Dynamic Logits Adjustment, and (c) Consistency-Guided Exploratory Cache.

	DLA	CGEC	Acc (%)
(a)	×	×	69.40
	✓	×	70.63
	✓	✓	**71.88**
	Entropy	**Logits**	**Acc (%)**
(b)	×	×	69.40
	✓	×	69.85
	×	✓	70.39
	✓	✓	**70.63**
	SCF	**TCF**	**Acc (%)**
(c)	×	×	70.63
	✓	×	71.26
	×	✓	71.37
	✓	✓	**71.88**

Table 4. Sensitivity analysis of α in Eq. 9 and β in Eq. 11 on the DTD dataset.

	$\beta = 0$	$\beta = 0.5$	$\beta = 1$	$\beta = 1.5$
$\alpha = 1.0$	56.68	57.62	57.10	57.34
$\alpha = 1.5$	56.58	57.45	58.51	57.62
$\alpha = 2.0$	58.51	**58.86**	58.82	58.27
$\alpha = 2.5$	57.82	57.85	57.92	57.80

racy to 71.88%, demonstrating that the consistency-guided exploratory cache provides complementary benefits by selecting more exploratory samples for adaptation.

Ablation on components of DLA. DLA influences prediction through entropy adjustment in Eq. 15 during training and logits adjustment in Eq. 14 at inference. We ablate these two components to evaluate their individual contributions. As shown in Tab. 3, entropy adjustment alone improves accuracy from 69.40% to 69.85%, while logits adjustment alone further improves it to 70.39%. Applying both yields the best performance of 70.63%, suggesting that they play complementary roles in enhancing accuracy and calibration.

Ablation on the components of CGEC. Tab. 3 further decomposes CGEC into the Semantic Consistency Filter (SCF) and the Temporal Consistency Filter (TCF). Starting from DLA without CGEC (70.63%), enabling SCF increases accuracy to 71.26%, while TCF gives a slightly higher 71.37%. Activating both filters leads to the highest accuracy of 71.88%, indicating that enforcing both semantic and temporal consistency is essential for filtering exploratory samples and achieving stable adaptation.

Parameter sensitivity analysis. Tab. 4 shows the sensitivity of DLAE to the hyperparameters α in Eq. 9 and β in Eq. 11 on DTD. Across all settings, the performance varies only slightly, from 56.58% to 58.86%, indicating that DLAE is robust to the choice of hyperparameters. The best result is achieved at $\alpha = 2.0$ and $\beta = 0.5$ (58.86%), while nearby settings also yield comparable performance, suggesting that DLAE does not require careful tuning.

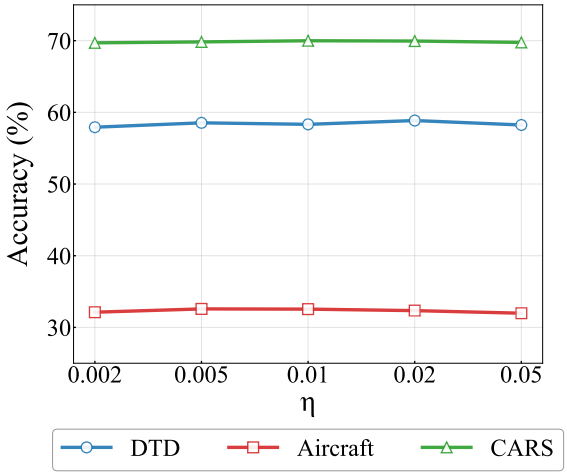


Figure 3. Sensitivity analysis of the parameter η, which is defined in Eq. 13. The figure shows how different values of η affect model performance across various datasets.

Fig. 3 shows the effect of η in Eq. 13, which controls the temporal decay of cached samples in TCF. Across datasets, accuracy stays stable as η varies from 0.002 to 0.05, with fluctuations below 1%. Moderate values (0.005–0.02) perform slightly better, indicating a trade-off between stability and adaptability. Overall, DLAE is robust to η, confirming the reliability of the temporal regularization mechanism.

5. Conclusion

We presented Dynamic Logits Adjustment and Exploration (DLAE), a unified framework for test-time adaptation of vision-language models under distribution shift. Unlike prior TTA methods that mainly rely on entropy-filtered high-confidence samples and thus inherit class-wise prediction biases and limited test coverage, DLAE jointly adjusts logits and explores uncertain regions of the test distribution. Dynamic Logit Adjustment (DLA) recalibrates logits using test-time prediction statistics, mitigating class-wise inconsistencies and improving recognition of weak or ambiguous samples. Built on these refined logits, the Consistency-Guided Exploratory Cache (CGEC) selectively targets boundary samples whose predictions are sensitive to logit adjustment, while enforcing semantic and temporal consistency for reliability. Extensive experiments across multiple benchmarks show that DLAE consistently surpasses state-of-the-art TTA methods in stability, adaptability, and generalization across diverse real-world applications and deployment scenarios.

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